RELIABLE FINANCIAL PLANNING WITH TEMPORAL MEMORY AND PLAN REPAIR

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1 PROBLEM DEFINITION AND MOTIVATION

Agentic AI is being increasingly pursued as a solution for personalized financial planning (Acharya et al., 2025; Bhat & Krishnan, 2025; Elgendy et al., 2025). A useful financial plan requires tracking income, payment dates, recurring obligations, and savings priorities as they change. Students and young professionals managing irregular income or upcoming purchases may need to revise these plans repeatedly. For example, a user might plan to save for a trip, later report that a paycheck is delayed, and then clarify that a rent increase takes effect next month. An assistant must distinguish current facts from future changes and corrections, and determine whether an existing plan remains feasible. However, research on long-context retrieval and conversational memory identifies difficulties recalling relevant information and maintaining accurate state (Liu et al., 2024; Kamradt, 2023; Wu et al., 2024), motivating an investigation of how memory design affects planning reliability.

While a rule-based system can verify results when its inputs and constraints are fully specified, conversational updates require interpretation, evidence retrieval, and contradiction resolution. Furthermore, we must maintain a financial plan depending on both the incoming update and the results of subsequent checks, thus these requirements motivate a tool-using agent architecture. We will build an agent that creates and revises budgets and savings schedules using transaction records and conversational updates. Our research questions are: **RQ 1:** How do memory representations affect an agent's ability to maintain valid plans as user information changes? **RQ 2:** Does checking recorded plan dependencies reduce unnecessary replanning without increasing the rate of invalid plans? We hypothesize that storing temporal validity and correction history will reduce outdated information errors, while recognizing that extraction mistakes may offset those benefits.

How our project fits within the scope of CS329Z: This project connects memory, retrieval, tool use, agent loops, and evaluation. A calculator or solver can check a fully specified budget, but the surrounding workflow requires interpreting updates, retrieving evidence, resolving contradictions, and revising decisions after feedback. We will use a single tool-using agent and a deterministic simulator to keep the focus on state management and planning reliability.

2 COMPARISON WITH EXISTING WORK

Financial Assistants: Prior work mostly focuses on evaluating status questions-and-answering scenarios: GPT-4 Turbo with retrieval answered incorrectly or refused 81% of questions on FinanceBench (Islam et al., 2023), while FinBen and ConvFinQA only test numerical reasoning over static documents (Xie et al., 2024b; Chen et al., 2022). No prior work focuses on the validity of an agent's financial planning as user's facts change.

Memory: Long-context models struggle with recall based on position and length of contexts (Liu et al., 2024; Kamradt, 2023; Hsieh et al., 2024). Seminal benchmarks such as LongMemEval (Wu et al., 2024) and LoCoMo (Maharana et al., 2024) test whether agents recall facts across long interactions, including knowledge updates and temporal reasoning, but only in general contexts. Seminal agent memory systems address storage but not temporal validity. MemGPT overwrites free-text memory (Packer et al., 2023), while Mem0 only updates or deletes facts in place, disregarding temporal history (Chhikara et al., 2025). Zep attaches validity intervals to facts in a temporal knowledge graph, improving this recall, but is only evaluated on question-answering, and not downstream decisions, and does not support unresolved contradictions (Rasmussen et al., 2025).

Planning and verification: LLM agents struggle with multiple constraints: ReAct with GPT-4 Turbo passes only 0.6% of TravelPlanner tasks (Yao et al., 2022; Xie et al., 2024a). LLMs cannot reliably self-verify (Huang et al., 2024; Stechly et al., 2025), and external critics raise the pass rate only to 20.6% (Kambhampati, 2024; Gundawar et al., 2024). These critics also trust the given specification, so a plan built on incorrectly recalled facts still passes. τ-bench reveals inconsistent agent reliability (Yao et al., 2024), but without evolving facts. For RQ2, truth maintenance systems revise only beliefs dependent on a retracted premise (Doyle, 1979), and plan repair is more stable than replanning (Fox et al., 2006). LLM replanners such as AdaPlanner (Sun et al., 2023) lack dependency records and so cannot tell when an update is irrelevant.

Our contributions: We combine three elements that prior work treats separately: memory that preserves temporal validity and correction history, plans that record the fact versions they depend on, and decision-level evaluation against hidden ground truth. This lets us measure whether failures arise from extraction, retrieval, temporal interpretation, or planning, and whether dependency-based repair avoids unnecessary revisions without admitting invalid plans.

3 Next Steps and Direction

System Design: The agent will use tools to retrieve transactions and obligations, request clarification, simulate cash flow, and save plans. Its memory will store each fact's value, source passage, observation time, effective interval, and status (including superseded or unresolved), so corrections preserve history. Plans will record the fact versions behind their constraints. Relevant updates will trigger validation and, when necessary, repair; unresolved contradictions will trigger clarification. A simulator will check dated cash balances, required payments, goal deadlines, and minimum reserves. It will return specific violations for a bounded repair loop. This verifier uses the agent's reconstructed state, while a separate evaluator uses hidden ground truth, so an internally consistent plan built on an incorrect fact can still fail.

Data: We will generate 200 synthetic episodes (50 development, 150 test), each with multiple interactions. A programmatic generator will define balances, dated income and bills, goals, and feasible or infeasible outcomes. Updates will cover corrections, delayed payments, future effective changes, goal pauses, conflicting statements, and irrelevant information, combined and varied in history length. The agent will learn facts only through observations or scripted user answers, never from the hidden state. If LLMs generate wording, numerical states and labels remain programmatically controlled. We will manually audit at least 25 episodes and keep profiles and update combinations separate across splits.

Evaluation: We will compare four memory methods: full history (within the context limit), rolling summaries, transcript retrieval, and structured temporal memory, each replanning after every update. A second experiment will compare full replanning with dependency-based repair using structured memory. At each update, the agent will return a plan, declare infeasibility or uncertainty, or exhaust its call budget. Code-based graders will check every required constraint and accept any valid plan; a reference solver will establish feasibility. Primary metrics are valid-plan completion on feasible requests, correct infeasibility detection, and outdated-fact errors. Secondary metrics are repair success, unnecessary revisions, latency, and token cost, with abstentions reported separately. Three runs per test episode will estimate consistency, with uncertainty intervals computed over episodes. Trace analysis will distinguish extraction, retrieval, temporal interpretation, and planning errors; an oracle-state diagnostic will help isolate memory related failures.

Milestones and Scope: We first build the simulator, graders, and a 25-episode pilot, aiming for an end-to-end prototype and initial memory comparisons by the midpoint. We will then expand the evaluation, test repair policies, and document failure modes. Deliverables will include a demo interface and a repository with reproducible evaluation commands. All accounts and actions will be simulated; the initial scope is budgeting and savings, with no real transfers or trades. Synthetic data supports reproducibility but limits conclusions about real financial behavior.

4 AI-Use Disclosure Statement

AI tools were used to brainstorm project ideas, research related work, and to edit text for concision.

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